

# Lab07\_Padovani

Rafael Padovani

## Laboratório 7

### SQLite Parte 2

```
library(RSQLite)
```

Warning: pacote 'RSQLite' foi compilado no R versão 4.6.1

1. Baixe o arquivo uwmadison.sqlite3. Conecte-se a ele usando o pacote RSQLite, armazenando a conexão em uma variável conn.

```
path = "Data"
fname = file.path(path, "database.sqlite3")
conn <- dbConnect(RSQLite::SQLite(), fname)
```

```
dbListTables(conn)
```

```
[1] "course_offerings"  "courses"           "grade_distributions"
[4] "instructors"       "rooms"              "schedules"
[7] "sections"          "subject_memberships" "subjects"
[10] "teachings"
```

```
dbListFields(conn, "teachings")
```

```
[1] "instructor_id" "section_uuid"
```

2. Quem são e quantos são os professores que lecionaram disciplinas cujo tópico era estatística (subjects.abbreviation='STAT')

```

dbGetQuery(conn, "
  SELECT COUNT(DISTINCT i.id) AS qnt_professores_stat
  FROM instructors i
  JOIN teachings t ON i.id = t.instructor_id
  JOIN sections se ON t.section_uuid = se.uuid
  JOIN course_offerings co ON se.course_offering_uuid = co.uuid
  JOIN subject_memberships sm ON co.uuid = sm.course_offering_uuid
  JOIN subjects su ON sm.subject_code = su.code
  WHERE su.abbreviation = 'STAT'
")

```

```

      qnt_professores_stat
1                379

```

```

dbGetQuery(conn, "
  SELECT DISTINCT i.name
  FROM instructors i
  JOIN teachings t ON i.id = t.instructor_id
  JOIN sections se ON t.section_uuid = se.uuid
  JOIN course_offerings co ON se.course_offering_uuid = co.uuid
  JOIN subject_memberships sm ON co.uuid = sm.course_offering_uuid
  JOIN subjects su ON sm.subject_code = su.code
  WHERE su.abbreviation = 'STAT'
  ORDER BY i.name
  LIMIT 10
")

```

```

      name
1      ABIGAIL BENZINE
2  ADIN-CRISTIAN ANDREI
3      AKICHIKA OZEKI
4          ALAN HUANG
5      ALBERTO DEL PIA
6      ALBRECHT KLEMM
7  ALEXANDER COVINGTON
8      ALEXANDER FISH
9      ALEXANDER KISELEV
10     ALYSSA DIGILIO

```

3. O GPA americano é definido numa escala de 0 a 4, em que A = 4, AB = 3.5, B = 3, BC = 2.5, C = 2, D = 1 e F = 0. Determinando a nota média de cada oferecimento pela

ponderação da quantidade de alunos em cada extrato com os valores numéricos de cada conceito, indique (no que se referente a disciplinas no assunto de estatística):

Quem é o professor mais difícil?

```
dbGetQuery(conn, "
  SELECT i.name AS professor,
         SUM(gd.a_count*4 + gd.ab_count*3.5 + gd.b_count*3 + gd.bc_count*2.5 + gd.c_count*2 + gd.d_count) /
         SUM(gd.a_count + gd.ab_count + gd.b_count + gd.bc_count + gd.c_count + gd.d_count) AS media_gpa
  FROM instructors i
  JOIN teachings t ON i.id = t.instructor_id
  JOIN sections se ON t.section_uuid = se.uuid
  JOIN grade_distributions gd ON se.course_offering_uuid = gd.course_offering_uuid
                           AND se.number = gd.section_number
  JOIN subject_memberships sm ON se.course_offering_uuid = sm.course_offering_uuid
  JOIN subjects su ON sm.subject_code = su.code
  WHERE su.abbreviation = 'STAT'
         AND (gd.a_count + gd.ab_count + gd.b_count + gd.bc_count + gd.c_count + gd.d_count + gd.f_count) > 0
  GROUP BY i.id, i.name
  ORDER BY media_gpa ASC
  LIMIT 1
")
```

|   | professor      | media_gpa |
|---|----------------|-----------|
| 1 | JAMES D KUELBS | 2.6375    |

Quem é o professor mais fácil?

```
dbGetQuery(conn, "
  SELECT i.name AS professor,
         SUM(gd.a_count*4 + gd.ab_count*3.5 + gd.b_count*3 + gd.bc_count*2.5 + gd.c_count*2 + gd.d_count) /
         SUM(gd.a_count + gd.ab_count + gd.b_count + gd.bc_count + gd.c_count + gd.d_count + gd.f_count) AS media_gpa
  FROM instructors i
  JOIN teachings t ON i.id = t.instructor_id
  JOIN sections se ON t.section_uuid = se.uuid
  JOIN grade_distributions gd ON se.course_offering_uuid = gd.course_offering_uuid
                           AND se.number = gd.section_number
  JOIN subject_memberships sm ON se.course_offering_uuid = sm.course_offering_uuid
  JOIN subjects su ON sm.subject_code = su.code
  WHERE su.abbreviation = 'STAT'
         AND (gd.a_count + gd.ab_count + gd.b_count + gd.bc_count + gd.c_count + gd.d_count + gd.f_count) > 0
  GROUP BY i.id, i.name
  ORDER BY media_gpa DESC
  LIMIT 1
")
```

```
ORDER BY media_gpa DESC
LIMIT 1
")
```

```

                                professor media_gpa
1 SRIKANTHMADHAVAN ARAVAMUTHAN                                4
```

Qual é a disciplina mais difícil?

```
dbGetQuery(conn, "
SELECT c.name AS disciplina,
       SUM(gd.a_count*4 + gd.ab_count*3.5 + gd.b_count*3 + gd.bc_count*2.5 + gd.c_count*2 +
          SUM(gd.a_count + gd.ab_count + gd.b_count + gd.bc_count + gd.c_count + gd.d_count +
FROM courses c
JOIN course_offerings co ON c.uuid = co.course_uuid
JOIN grade_distributions gd ON co.uuid = gd.course_offering_uuid
JOIN subject_memberships sm ON co.uuid = sm.course_offering_uuid
JOIN subjects su ON sm.subject_code = su.code
WHERE su.abbreviation = 'STAT'
      AND (gd.a_count + gd.ab_count + gd.b_count + gd.bc_count + gd.c_count + gd.d_count + gd.
GROUP BY c.uuid, c.name
ORDER BY media_gpa ASC
LIMIT 1
")
```

```

                                disciplina media_gpa
1 Introduction to the Theory of Probability 2.886374
```

Qual é a disciplina mais fácil?

```
dbGetQuery(conn, "
SELECT c.name AS disciplina,
       SUM(gd.a_count*4 + gd.ab_count*3.5 + gd.b_count*3 + gd.bc_count*2.5 + gd.c_count*2 +
          SUM(gd.a_count + gd.ab_count + gd.b_count + gd.bc_count + gd.c_count + gd.d_count +
FROM courses c
JOIN course_offerings co ON c.uuid = co.course_uuid
JOIN grade_distributions gd ON co.uuid = gd.course_offering_uuid
JOIN subject_memberships sm ON co.uuid = sm.course_offering_uuid
JOIN subjects su ON sm.subject_code = su.code
WHERE su.abbreviation = 'STAT'
```

```

        AND (gd.a_count + gd.ab_count + gd.b_count + gd.bc_count + gd.c_count + gd.d_count + gd.e_count) > 0
    GROUP BY c.uuid, c.name
    ORDER BY media_gpa DESC
    LIMIT 1
")

```

```

                                disciplina media_gpa
1 Nonparametric Statistics and Machine Learning Methods                4

```

Desconecte do banco de dados.

```
dbDisconnect(conn)
```

```

# Exibindo o horário de execução do código
Sys.setenv(TZ = "America/Sao_Paulo")
Sys.time()

```

```
[1] "2026-09-22 01:29:19 -03"
```